

1. **DEFINITION**

DEFINITION
Glyceryl dibehenate E 471/GRAS/FCC

2. **FIELD OF USE**

FIELD OF USE
This ingredient must be used according to appropriate regulations
Nutraceutical ingredient

3. **MANUFACTURING PROCESS AND COMPOSITION**

MANUFACTURING PROCESS
Esterification reaction between glycerol and behenic acid. The esterification step is followed by atomization using a spray-cooling technique. No solvent or catalyst used during the manufacturing process.

COMPOSITION
Well defined multi-constituent substance composed of mono-, di- and triglycerides of Behenic acid (C22), the diester fraction being predominant (Specification: Mono and diesters > 70%).
Full Labeling (Key letters system)
$A_1 \geq 75.0$; $50.0 \leq A_2 < 75.0$; $25.0 \leq B < 50.0$; $10.0 \leq C < 25.0$; $5.0 \leq D < 10.0$; $1.0 \leq E < 5.0$; $0.1 \leq F < 1.0$; $G < 0.1$
Glyceryl behenate
$A_1 \geq 75.0\%$
Other constituents (i.e. incidental ingredients)
None



Regulatory data sheet (RDS) – Nutraceutical market

COMPRITOL E ATO (Code: 7101)

Last update: 24 September 2020

4. FOOD/NUTRACEUTICAL/DIETARY SUPPLEMENT APPLICATIONS

Mono- and Diglycerides have a long history of use worldwide as food additives for use in food, nutraceutical products and dietary supplements.

COMPRITOL E ATO is authorised for use as food additive in nutraceutical products and dietary supplements.

CONFORMITY TO FOOD ADDITIVE REGULATION			
THE INGREDIENT IS APPROVED FOR USE IN NUTRACEUTICAL PRODUCTS / DIETARY SUPPLEMENTS IN			DETAILS ON FOOD ADDITIVE STATUS
EU	US	Japan	
✓	✓		E 471 / GRAS / FCC
FCC: Food chemical codex ; GRAS: Generally recognized as safe ; USFA : US food additive ; USIFA: US indirect food additive ; E xxx: European food additive ; JSFA : Japanese standard food additive			

Country	Reference	Registration number
International	Evaluations Performed by Joint FAO/WHO Expert Committee on Food Additives	INS471 Mono- and Diglycerides
U.S.A.	FCC (Food Chemical Codex)	Mono- and Diglycerides
U.S.A.	GRAS US Food additives according to 21 CFR	21 CFR §184.1505 Mono- and Diglycerides

5. SPECIFIC REGULATIONS

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - Australian Register of Therapeutic Goods Ingredients			
Status	Ingredient Name	Use	Restrictions
Approved	Glyceryl behenate Synonyms: 2,3-Dihydroxypropyl docosanoate Glycerol dibehenate (EP)	Available for use as an Active Ingredient in: Biologicals, Prescription Medicines Available for use as an Excipient Ingredient in: Biologicals, Devices, Export Only, Listed Medicines, Over the Counter, Prescription Medicines Not available as an Equivalent Ingredient in any application	<u>Applies to over the counter</u> Glyceryl behenate is: Glyceryl behenate is i) permitted for use as an oral excipient; ii) permitted for use as a lubricant for tablets and capsules at a final concentration of 1-3%; and iii) permitted for use as a slow release lipophilic matrix for sustained release tablets and capsules at a final concentration of 10-30%. (up to a maximum of 500mg/person/day) Glyceryl behenate is approved for use as an excipient ingredient for topical use at a concentration of 5% or less. <u>Applies to listed medicines:</u> Behenic acid is a mandatory component of glyceryl behenate. When for oral ingestion, the maximum recommended daily dose must not provide more than 383.5 milligrams of behenic acid. In medicines for topical use, the concentration of glyceryl behenate must be no more than 5%.



Regulatory data sheet (RDS) – Nutraceutical market

COMPRITOL E ATO (Code: 7101)

Last update: 24 September 2020

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - [Australian Register of Therapeutic Goods Ingredients](#)

Status	Ingredient Name	Use	Restrictions
Approved	Glycerol dibehenate	Available for use as an Active Ingredient in: Biologicals, Prescription Medicines Available for use as an Excipient Ingredient in: Biologicals, Export Only, Over the Counter, Prescription Medicines Not available as an Equivalent Ingredient in any application	/

AUSTRALIA: Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homoeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.

6. CHEMICAL REGULATIONS

CHEMICAL REGULATION

INCI NAME	CAS #	EINECS #	Chemical name
Glyceryl behenate	77538-19-3 (or 91052-55-0) (or 30233-64-8) (or 94201-62-4) (or 18641-57-1)	278-717-5 (or 293-216-1) (or 250-097-0) (or 303-650-6) (or 242-471-7)	Docosanoic acid, ester with 1,2,3-propanetriol Glycerides, C16-22 mono-, di- and tri- Docosanoic acid, monoester with 1,2,3-propanetriol 2-hydroxypropane-1,3-diyl didocosanoate Docosanoic acid, 1,2,3-propanetriyl ester

NATIONAL INVENTORIES

INCI NAME	U.S.A. (TSCA)	Canada (DSL/NDSL)	Australia (AICS)	China (IECSC)	Other inventories
Glyceryl behenate	Listed	Listed	Listed	Listed	Docosanoic acid, ester with 1,2,3-propanetriol: EINECS, ASIA-PAC, NZIoC, TSCI, IECSC (or Glycerides, C16-22 mono-, di- and tri-: EINECS, PICCS, ASIA-PAC, ECL serial n°KE-17933, IECSC) (or Docosanoic acid, monoester with 1,2,3-propanetriol: TSCA, NDSL, AICS, ASIA-PAC, NZIoC, EINECS, ECL serial n°KE-12785, R-ICL, IECSC, TCSI) (or 2-hydroxypropane-1,3-diyl didocosanoate: EINECS, ASIA-PAC, IECSC, R-ICL) (or Propane-1,2,3-triyl tridocosanoate: DSL, AICS, PICCS, ASIA-PAC, NZIoC, IECSC, TCSI, EINECS)

U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics ... are not subject to TSCA regulation.

Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer. In addition, substances not listed in DSL or NDSL, but listed in the "in-commerce list" can be freely used in Products Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as "Natural" according Health Canada's definition are exempted from New Substances regulation.

Australia: Substances Introduced at 1% or less in cosmetics are exempted from AICS listing. Otherwise, substances not listed on AICS can be imported in Australia in quantities not exceeding 10 kg per 12-month period and per importer. In addition, in the Industrial Chemicals (Notification and Assessment) Act 1989 (the Act), an industrial chemical which meets the definition of 'a naturally-occurring chemical' is taken to be on the Australian Inventory of Chemical Substances (AICS), whether listed or not, and therefore does not require notification as a new chemical prior to manufacture or importation into Australia.

Regulatory data sheet (RDS) – Nutraceutical market

COMPRITOL E ATO (Code: 7101)

Last update: 24 September 2020

REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
INCI NAME	Status of the product	Status per substance
Glyceryl behenate	Exempted	Substances intended to be used in food products are exempted from the main provisions on registration, evaluation and authorization, under REACH regulation.

Liability clause

The information contained in this document is based on our knowledge of the related product(s) at the date indicated herein, and is provided in good faith. Notwithstanding the foregoing, no guarantee is made in respect of the product's(s') properties or efficacy, or the continued exactitude and completeness of the information herein. No contractual tie is formed by the provision of this information. This document does not relieve the user from conducting all tests and verifications to ensure the product's(s') suitability for each intended use. Nothing herein shall be construed as a recommendation or licence to use any information that does or may conflict with any intellectual property. We make no representation or warranty, express or implied, that use of this information will not infringe any intellectual property of Gattefossé or any third party. The user assumes all risks and liability incurred if a product is used other than for the purpose designated by Gattefossé. This statement does not relieve the user from verifying its conformity with all current applicable laws and regulations, and the user shall remain solely liable for ensuring its continued conformity therewith.

